Design Proposal for Group-Index Extraction Using a Silicon Photonic Mach–Zehnder Interferometer

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Abstract—This proposal describes a silicon photonic Mach–Zehnder interferometer (MZI) for extracting quasi-TE waveguide group index from free spectral range (FSR). The revised device uses a 500 nm × 220 nm silicon strip waveguide, one TE_1550 input grating coupler, two TE_1550 outputs, a Y-branch input splitter, and a broadband two-output combiner. The layout targets a 100 µm arm-length imbalance. Lumerical MODE gives n_eff = 2.444580 and n_g = 4.201897 at 1550 nm; the resulting predicted FSR is 5.718 nm. A SiEPIC-extracted EBeam compact-model circuit was simulated in INTERCONNECT; two approximately complementary output spectra were obtained. The MODE compact-model fit spans 1550–1575 nm, while the INTERCONNECT spectrum spans approximately 1500–1600 nm.

I. INTRODUCTION

Silicon waveguide dispersion affects the relative optical delay in photonic interferometers. The group index, n_g, describes how propagation phase changes with optical frequency and can be inferred from the free spectral range of an unbalanced MZI. The purpose of this proposal is to design a compact device for extracting n_g near 1550 nm and comparing the experimentally extracted value with a Lumerical MODE eigenmode calculation [1].

Only one device is proposed for fabrication: the revised $\Delta L \approx 100$ µm MZI. The ∼50 nm grating-coupler-limited test window and 1 pm minimum wavelength step specified by the course should permit several fringes to be resolved.

II. THEORY

The propagation constant of a guided mode with effective index n_eff is

$$\beta(\lambda) = 2\pi\, n_{eff}(\lambda) / \lambda. \qquad (1)$$

For unequal interferometer arms with the same nominal waveguide cross section, the differential propagation phase is

$$\Delta\varphi(\lambda) \approx \beta(\lambda)\, \Delta L + \varphi_c(\lambda), \quad \Delta L = L_{long} - L_{short}. \qquad (2)$$

The term φ_c captures any additional wavelength-dependent differential phase from bends and couplers. For an ideal balanced Y-branch and a nominal 3 dB two-output combiner, the normalized transfer functions are

$$T_A(\lambda) = \tfrac{1}{2}[1 + \cos \Delta\varphi(\lambda)], \quad T_B(\lambda) = \tfrac{1}{2}[1 - \cos \Delta\varphi(\lambda)]. \qquad (3)$$

These are idealized complementary outputs; actual fringe visibility and insertion loss depend on the component models. The waveguide group index is

$$n_g(\lambda) = n_{eff}(\lambda) - \lambda\, [d\, n_{eff}(\lambda) / d\lambda]. \qquad (4)$$

A 2π change of differential phase between adjacent fringes gives the local FSR when the extra coupler phase dispersion is negligible:

$$FSR(\lambda) \approx \lambda^2 / [n_g(\lambda)\, \Delta L]. \qquad (5)$$

Therefore, adjacent measured transmission maxima at λ_m and λ_(m+1) give

$$n_{g,exp} \approx \lambda_{mid}^2 / [\Delta L(\lambda_{(m+1)} - \lambda_m)]. \qquad (6)$$

Here λ_mid = (λ_m + λ_(m+1))/2. Several adjacent peak separations will be used to test consistency and reduce peak-picking uncertainty.

III. MODELING AND SIMULATION

A. Waveguide geometry and eigenmode simulation

Simulation Study 1 (MODE): The 500 nm × 220 nm silicon strip waveguide was analyzed at 1550 nm. For the fundamental quasi-TE mode, MODE returned Re(n_eff) = 2.444580, n_g = 4.201897, and TE polarization fraction = 98%. The waveguide intensity profile appears in Fig. 1. A tracked frequency sweep produced 101 numerical samples from 1550 to 1575 nm, shown in Fig. 2.

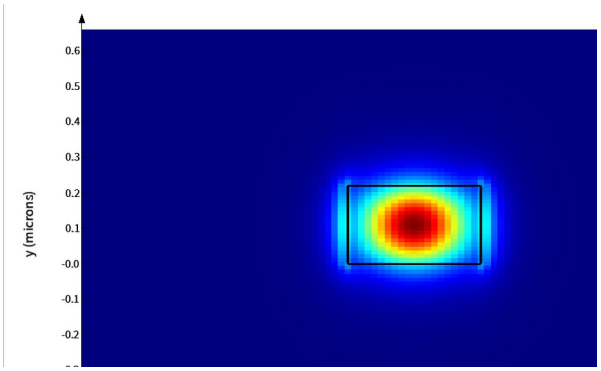


Fig. 1. Lumerical MODE fundamental quasi-TE electric-field intensity (normalized), at λ = 1550 nm, for the 500 nm × 220 nm silicon strip waveguide.

The exported effective-index data and corresponding group-index values are plotted in Fig. 2. The 1550–1575 nm sweep characterizes the dispersion near the nominal 1550 nm operating point; the proposed grating-coupler measurement window is approximately 1525–1575 nm. No TM simulation was required because the submitted circuit uses quasi-TE grating couplers.

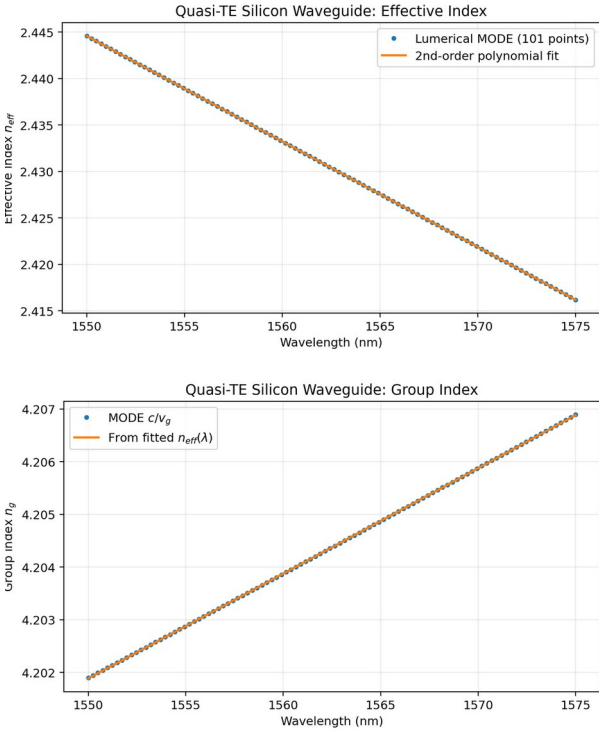


Fig. 2. Effective and group indices of the quasi-TE waveguide (1550–1575 nm): discrete Lumerical MODE outputs and second-order polynomial representation based on 101 exported points.

The wavelength-dependent real effective index was fitted by second-order least squares about λ_0 = 1.55 µm (wavelength expressed in µm):

$$n_{eff}(\lambda) = 2.44457983 - 1.13374593(\lambda - 1.55) - 0.06400363(\lambda - 1.55)^2. \quad (7)$$

The fitted coefficients are a_0 = 2.44457983, a_1 = −1.13374593 µm^{-1} and a_2 = −0.06400363 µm^{-2}. Across 101 MODE samples (1550–1575 nm), the root-mean-square fitting error is 5.18 × 10^{-9} and the maximum error is 1.34 × 10^{-8}. The fitted derivative yields n_g(1550 nm) = 4.201886, compared with the MODE-reported 4.201897.

B. Mach–Zehnder interferometer design

Fig. 3. Schematic of the proposed MZI: input GC1, Y-branch, unequal arms, broadband combiner and two measured outputs GC2/GC3; not drawn to physical scale.

The existing long-arm U-turn was shifted from x = 95.130 µm to x = 52.792 µm at two path vertices without moving the couplers or other connections. Using the earlier estimated 184.677 µm imbalance, this gives ΔL_new ≈ 184.677 − 2(95.130 − 52.792) = 100.001 µm. The final waveguide lengths remain subject to extraction from the saved GDS; a KLayout/SiEPIC functional layout check reported no errors.

Design feature	**This design**
Waveguide / polarization	500 × 220 nm Si strip / quasi-TE
Input splitter	Y-branch
Output combiner	Broadband directional coupler
Optical I/O	1 input GC + 2 output GCs
GC pitch	127 µm between adjacent GCs
Arm length difference	ΔL ≈ 100 µm (target)
Center wavelength	1550 nm
Nominal test range	Approx. 1525–1575 nm

Using the simulated MODE group index n_g(1550 nm) = 4.201897, the predicted free spectral range of the target device is

$$FSR(1550\ nm) \approx 1.55^2 / (4.201897 \times 100)\ \mu m = 5.7177\ nm. \quad (8)$$

An ideal 50 nm usable grating-coupler window would contain approximately 50/5.718 ≈ 8.7 fringe periods. The actual usable bandwidth and fringe visibility depend on grating-coupler and splitter/combiner response. The expression neglects additional wavelength-dependent component-phase contributions.

Table I is an analytical sensitivity comparison using MODE n_g(1550 nm) = 4.201897. Only ΔL ≈ 100 µm represents the revised GDS; other imbalances are hypothetical and do not imply additional fabricated devices.

ΔL (µm)	FSR at MODE n_g (nm)	**Status**
75	7.624	Theory only
100	5.718	Actual layout target
125	4.574	Theory only
150	3.812	Theory only

Simulation Study 2 (INTERCONNECT): The finalized KLayout MZI GDS was exported using the SiEPIC circuit-simulation workflow. Its EBeam compact-model netlist was loaded into INTERCONNECT, with an optical network analyzer (ONA) automatically connected to the input grating-coupler channel and both output detection channels. The TE gain (dB) was swept over approximately 1500–1600 nm. This circuit simulation uses installed EBeam library models, rather than a direct full-wave electromagnetic simulation of the GDS; the MODE-fitted polynomial was analyzed separately.

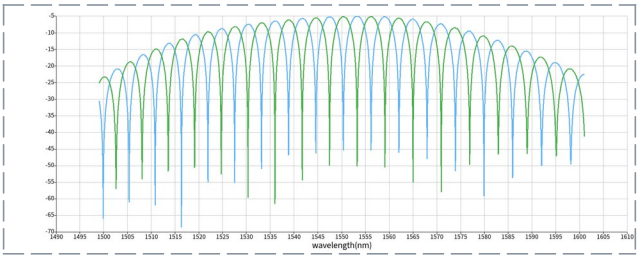


Fig. 4. SiEPIC-extracted MZI circuit simulated in Lumerical INTERCONNECT: TE transmission gain (dB) versus wavelength for GC1 to the two output channels. Blue: ONA input 1 (detector 1); green: ONA input 2 (detector 2). Source: student INTERCONNECT output. The nominal GDS arm imbalance is ΔL ≈ 100 µm.

The two simulated outputs show complementary fringes within a wavelength-dependent grating-coupler envelope. From the image, adjacent maxima near 1550 nm are separated by roughly 5.7–5.8 nm, consistent with the MODE-based prediction of 5.718 nm. Using an illustrative image-estimated FSR of 5.78 nm gives n_g,FSR ≈ 24.025 / 5.78 ≈ 4.16, compared with n_g,MODE = 4.201897. This is an approximate visual estimate, not a measurement from exported INTERCONNECT data. Coupler phase and imperfect ΔL may also contribute.

IV. FABRICATION AND LAYOUT

The revised GDS uses the course filename EBeam_ruisige.gds. The proposed three-coupler geometry preserves the 127 µm fibre-array pitch, with the longer arm shortened to target ΔL ≈ 100 µm. KLayout/SiEPIC functional verification displayed “No layout

errors detected." Course-side automated verification, final repository merge into EBeam.gds, and measurement coordinates must be confirmed separately; successful local checks are not fabrication confirmation.

Optional presentation improvement: include a KLayout screenshot of the updated three-grating-coupler GDS, showing the shortened long arm and the device label. This is a layout illustration, not a separate optical simulation.

V. EXPERIMENTAL DATA (TO BE COMPLETED)

After fabrication, measure GC2 and GC3 transmission with GC1 as the input. Record the wavelength axis, wavelength step, test polarization, output power, extracted adjacent fringe wavelengths, local FSRs and measurement repeatability. Add a measured transmission spectrum here when data are supplied.

VI. ANALYSIS (TO BE COMPLETED)

After fabrication, determine experimental FSR from adjacent measured maxima and use Eq. (6) to extract n_g,exp; then compare with the MODE value of 4.201897 and the simulated complementary fringes in Fig. 4. Consider uncertainty from physical ΔL, peak finding, wavelength calibration, and possible wavelength-dependent extra phase of couplers and bends.

VII. CONCLUSION

This proposal presents one quasi-TE, 500 nm × 220 nm strip-waveguide MZI with nominal $\Delta L \approx 100$ μm and predicted FSR 5.718 nm at 1550 nm. Lumerical MODE produced the TE mode profile, dispersion curves, and a second-order waveguide compact model. SiEPIC-extracted INTERCONNECT simulation of the GDS produced two complementary transmission spectra with approximately 5.7–5.8 nm fringe spacing near 1550 nm, consistent with the theoretical prediction. Following fabrication, the measured spectra will be used to extract and compare experimental waveguide group index. Final repository submission and automated verification status must be checked separately.

REFERENCES

[1] L. Chrostowski and M. Hochberg, Silicon Photonics Design: From Devices to Systems. Cambridge University Press, 2015.

[2] UBCx Phot1x course instructions, "Design Proposal – Report" and "Final Layout Submission," 2026.

[3] Ansys Optics, "Modal Analysis Tab," documentation.

[4] Ansys Optics, "Optical Network Analyzer (ONA)," INTERCONNECT documentation.